

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	17 July 2026
Team ID	-----
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Dataset Collection	USN-1	As a data scientist, I can collect historical flood and weather datasets from reliable sources so that the model can be trained effectively.	3	High	Team	15 june 2026	21 juner 2026
Sprint-1	Data Exploration	USN-2	As a data analyst, I can perform Exploratory Data Analysis (EDA) to understand patterns, correlations, and missing values in the dataset.	3	High	Team	15 june 2026	21 june 2026
Sprint-1	Data Preprocessing	USN-3	As a data scientist, I can clean the dataset, handle missing values, encode features, and normalize data for model	5	High	Team	15 june	21 june 2026

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
			training.				2026	
Sprint-2	Model Development	USN-4	As a machine learning engineer, I can train Decision Tree, Random Forest, KNN, and XGBoost models to identify the best-performing algorithm.	8	High	Team	22 june 2026	31 june 2026
Sprint-2	Model Evaluation	USN-5	As a machine learning engineer, I can evaluate the models using Accuracy, Precision, Recall, F1-Score, ROC Curve, and Confusion Matrix.	5	High	Team	22 june 2026	31 june 2026
Sprint-2	Model Management	USN-6	As an administrator, I can save the trained XGBoost model and preprocessing scaler for deployment and future predictions.	2	High	Team	22 june 2026	31 june 2026
Sprint-3	Web Application Development	USN-7	As a user, I can access a web application to enter rainfall and weather parameters for flood prediction.	5	High	Team	01 july 2026	10 july 2026
Sprint-3	Flood Prediction	USN-8	As a user, I can receive a prediction indicating Flood Risk or No Flood Risk based on the provided weather parameters.	3	High	Team	01 july 2026	10 july 2026
Sprint-3	User Interface	USN-9	As a user, I can view prediction results through a simple, responsive, and user-friendly interface.	3	Medium	Team	01 july 2026	10 july 2026
Sprint-4	Deployment	USN-10	As an administrator, I can deploy the trained XGBoost model using a Flask web application for real-time predictions.	5	High	Team	11 july 2026	20 july 2026
Sprint-4	Testing & Validation	USN-11	As a tester, I can validate prediction accuracy, system performance, and application functionality before final	3	High	Team	11 july	20 july 2026

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
			deployment.				2026	
Sprint-4	Maintenance & Enhancement	USN-12	As an administrator, I can update datasets, retrain the model, and enhance the application with live weather APIs, GIS mapping, and SMS alerts in future releases.	5	Medium	Team	11 july 2026	20 july 2026

Total Story Points by Sprint

Sprint	Total Story Points
Sprint-1	11
Sprint-2	15
Sprint-3	11
Sprint-4	13